WW-P-351a
March 6, 1963

SUPERSEDING
Fed. Spec. WW-P-351
July 22, 1930
(See 6.8).

FEDERAL SPECIFICATION

PIPE; RED BRASS, (COPPER ALLOY NO. 230), SEAMLESS STANDARD PIPE SIZE, REGULAR AND EXTRA-STRONG

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The specification covers seamless red brass, 85% (copper alloy No. 230) pipe in all standard pipe sizes.

1.2 Classification. Brass pipe shall be of the following classes, as specified (6.2):

Class 1.—Regular.

Class 2.—Extra strong.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specification and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian agencies).
Fed. Std. No. 146—Tolerances for Copper and Copper Base Alloy Mill Products.
Fed. Test Method Std. No. 151—Metals, Test Methods.
Fed. Std. No. 185—Continuous Identification Marking of Copper and Copper Base Alloys.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:
MIL-C-3993—Copper and Copper Base Alloy Mill Products; Packaging of.

Military Standard:
MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:22Z
Check the source to verify that this is the current version before use.

procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 **Chemical requirements.** The material shall conform to the chemical requirements shown in table I.

TABLE I.—*Chemical requirements*

Copper alloy number	Element	Percent
230	Copper	83.0 to 86.0
	Lead	0.06 Max.
	Iron	.05 Max.
	Zinc	Remainder

3.2 **Temper.** Unless otherwise specified (6.2), the pipe shall be furnished in the annealed condition. When specified (6.2), pipe shall be furnished in the drawn temper.

3.3 **Mechanical properties.**

3.3.1 *Hydrostatic pressure.* The pipe shall withstand the hydrostatic test pressure specified in table II, when tested as specified in 4.4.1 (see 6.3).

3.3.2 *Bending.* Annealed pipe 2 inches and under in outside diameter shall withstand the bend test specified in 4.4.2.

3.3.3 *Expansion.* The pipe shall withstand the expansion test specified in 4.4.3, or, as an alternative, the flattening test specified in 4.4.3.1.

TABLE II.—*Hydrostatic-test pressure*

Standard pipe size	Wall thickness (nominal)	Internal hydrostatic test pressure (minimum)
	Class 1—Regular	
Inches	*Inch*	*P.s.i.*
1/8	0.062	1,000
1/4	.082	1,000
3/8	.090	1,000
1/2	.107	1,000
3/4	.114	1,000
1	.126	1,000
1-1/4	.146	1,000
1-1/2	.150	1,000
2	.156	1,000
2-1/2	.187	1,000
3	.219	1,000
3-1/2	.250	1,000
4	.250	900
5	.250	700
6	.250	600
8	.312	550
10	.365	550
12	.375	450
	Class 2—Extra strong	
1/8	0.100	1,000
1/4	.123	1,000
3/8	.127	1,000
1/2	.149	1,000
3/4	.157	1,000
1	.182	1,000
1-1/4	.194	1,000
1-1/2	.203	1,000
2	.221	1,000
2-1/2	.280	1,000
3	.304	1,000
3-1/2	.321	1,000
4	.341	1,000
5	.375	1,000
6	.437	1,000
8	.500	900
10	.500	700

3.3.4 *Tension test.* When specified (6.2), annealed pipe shall have a minimum tensile strength of 40,000 pounds per square inch (p.s.i.) and a minimum yield strength of 12,000 p.s.i. at 0.5 percent extension under load, when tested as specified in 4.4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:22Z
Check the source to verify that this is the current version before use.

3.3.5 *Residual stress.* Brass pipe shall be free from stress insofar as can be determined when tested as specified in 4.4.5.

3.3.6 *Microstructure (annealed pipe).* Annealed pipe shall be fully recrystallized with an average grain size not in excess of 0.050 millimeter (mm.) when determined as specified in 4.4.6.

3.4 **Tolerance.** Tolerance shall not exceed that specified in the following sections of Fed. Std. No. 146:

(*a*) Dimensions and weights, standard pipe sizes (nonrefractory alloys).

(*b*) Weight and wall thickness tolerances (nonrefractory alloys).

(*c*) Length tolerances (nonrefractory alloys).

(*d*) Squareness of cut (nonrefractory alloys).

(*e*) Roundness tolerances (drawn temper nonrefractory alloys).

3.4.1 *Roundness (annealed pipe).* The difference between the major and minor outside dimensions at any cross section of annealed pipe, when inspected at the mill, shall not exceed the percentages of the nominal diameter specified in table III.

TABLE III.—*Roundness of annealed pipe*

Pipe	Percentage of nominal outside diameter
5.562 inches and less outside diameter	3
Over 5.562 inches outside diameter	6

3.5 **Identification marking.** Unless otherwise specified (6.2), identification marking shall be in accordance with Fed. Std. No. 185.

3.6 **Workmanship.** The pipe shall be clean, free from scale, smooth, round, straight, and of the proper dimensions. It shall be free from injurious grooving, indentations, cracks, flaws, slivers, spilly metal, or other harmful defects which would interfere with the use of the material in the application for which it is intended.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Supplier's responsibilities for inspection.** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling for quality conformance inspection.**

4.2.1 *Inspection lot.* For the purpose of inspection, a lot shall consist of all pipe of the same class and size in the quantities shown in table IV.

TABLE IV.—*Inspection lot*

Nominal pipe size	Weight of lot
Inches	*Pounds*
Up to 1-1/2, incl.	5,000
Over 1-1/2 to 4, incl.	10,000
Over 4	20,000

4.2.2 Samples selected for purposes of the tests prescribed in this specification shall be selected in a manner as to represent correctly the material furnished. Samples for each destructive test shall be selected in accordance with table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:22Z
Check the source to verify that this is the current version before use.

TABLE V.—*Sample for destructive tests*

Number of pieces in lot	Number of sample pieces to be taken
1 to 50	1
51 to 200	2
201 to 1500	3
	0.2 percent of total number of pieces in the lot but not to exceed 10 sample pieces

4.2.3 *Sampling for examination.*

4.2.3.1 *Visual examination.* From each inspection lot, a representative sample of pipe shall be selected in accordance with table VI and shall be visually examined as specified in 4.3.1.

4.2.3.2 *Dimensional examination.* From each inspection lot, a representative sample of pipe shall be selected in accordance with table VI and shall be measured as specified in 4.3.2. The sample selected for dimensional examination may be the same as those selected for visual examination.

TABLE VI.—*Sampling for visual and dimensional examination AQL (approx.) =1.5 percent defective*

Number of pipes in an inspection lot	Number of pipes in sample	Acceptance number (defectives)	Rejection number (defectives)
40 or under ..	15	0	1
41 to 110	25	1	2
111 to 300	35	1	2
301 to 500	50	2	3
501 to 800	75	3	4
801 to 1300 ...	110	4	5
1301 and over ..	150	5	6

4.2.4 *Sampling for chemical analysis.* Two ounces of clean chips shall be taken from each sample selected as specified in 4.2.2 for chemical analysis.

4.3 Examination.

4.3.1 *Visual.* Each sample pipe selected in accordance with table VI shall be visually examined to verify conformance with this specification.

4.3.2 *Dimensional.* Each sample pipe selected in accordance with table VI shall be measured to verify conformance with this specification.

4.3.3 Any nonconforming pipe in an inspection sample shall not be offered for delivery, and if the number of nonconforming pipes in the sample exceeds the acceptance number specified in table VI for that sample size, this shall be cause for rejection of the entire lot. The disposition of nonconforming product shall be in accordance with MIL-STD-105.

4.3.4 *Examination of preparation for delivery.* The packaging, packing, and marking shall be examined to determine compliance with the requirements of section 5 of this specification.

4.4 Test procedures.

4.4.1 *Hydrostatic-pressure test.* Each pipe shall withstand an internal hydrostatic pressure which will subject the metal to a fiber stress of 8,000 pounds per square inch, the test pressure being that shown in table II. In no case will a test pressure of over 1,000 pounds per square inch be required. The pipe shall be perfectly tight under pressure and show no signs of bulging, cracks, flaws, porous places, or other defects. If any pipe fails the test, the pipe shall not be offered for delivery.

4.4.1.1 In lieu of the hydrostatic-pressure test specified in 4.4.1, the pipe may be tested under water with air at 60 to 100 pounds per square inch internal pressure.

4.4.2 *Bending test.* Annealed full sections of pipe, 2 inches and under in out-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:22Z
Check the source to verify that this is the current version before use.

side diameter, selected in accordance with 4.2.2 shall be bent cold 180 degrees around a pin the diameter of which is one and one-half times the inside diameter of the pipe, without cracking on the outside of the bent portion.

4.4.3 *Expansion test.* Specimens selected as specified in 4.2.2 shall, after annealing, withstand an expansion of 25 percent of the inside diameter when expanded by a tapered pin having a 60-degree angle (3.3.3), and the expanded area shall show no cracks or ruptures visible to the unaided eye.

4.4.3.1 *Flattening test.* In lieu of the test specified in 4.4.1, for pipe over 4 inches in diameter in the annealed condition, a section 4 inches in length shall be selected as specified in 4.2.2. The specimen shall be flattened so that a gage set at three times the wall thickness will pass over the pipe freely throughout the flattened part. The pipe so tested shall develop no cracks or ruptures visible to the unaided eye. In making this test, the elements shall be slowly flattened by one stroke of the press.

4.4.4 *Tensile test.* When required (3.3.4), tensile test specimens shall be selected as specified in 4.2.2. Specimens shall be full sections when practicable. Tensile tests and specimens shall be in accordance with Fed. Test Method Std. No. 151.

4.4.5 *Stress tests.* Full-section test specimen, 6 inches in length, shall be selected as specified in 4.2.2 and tested in accordance with method 831 of Fed. Test Method Std. No. 151.

4.4.6 *Microscopic examination.* The test specimens selected in accordance with 4.2.2 for microscopic examination shall be radial longitudinal sections of the pipe made by cutting along a plane containing the longitudinal axis of the pipe. Each of the specimens shall be examined in accordance with method 312 of Fed. Test Method Std. No. 151. Grain size shall be determined by direct comparison with standard micrographs. In case of dispute, the diameter of the average grain shall be determined by the Jeffries Planimetric Method.

4.4.7 *Chemical analysis.* Chemical analysis shall be made by the wet chemical method in accordance with method 111 of Fed. Test Method Std. No. 151, or by the spectrochemical method in accordance with method 112 of Fed. Test Method Std. No. 151. In case of dispute, the chemical analysis by wet chemical methods shall be the basis for acceptance.

4.5 **Rejection.** If any specimen fails to conform to this specification, this shall be cause for rejection of the entire lot, except 4.3.3 and 4.4.1, subject to retest provisions in accordance with Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 **Packaging.** Packaging shall be level A or C, as specified (6.2).

5.1.1 *Level A.* Packaging of piping shall be in accordance with level A of MIL-C-3993.

5.1.2 *Level C.* Packaging shall be in accordance with the supplier's commercial practice.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (6.2).

5.2.1 *Level A.* Piping, packaged as specified (5.1), shall be packed in accordance with level A requirements of MIL-C-3993.

5.2.2 *Level B.* Piping, packaged as specified (5.1), shall be packed in accordance with level B requirements of MIL-C-3993.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:22Z
Check the source to verify that this is the current version before use.

5.2.3 *Level C.* Piping, packaged as specified (5.1), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers or method of packing shall be in accordance with the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-C-3993.

6. NOTES

6.1 **Intended use.** Red brass pipe is intended for use where a material with resistance to corrosion by salt air or gases is required.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Class required (1.2).

(*c*) Temper, if other than annealed (3.2).

(*d*) If tension test is required (3.3.4).

(*e*) Pipe size required (3.4).

(*f*) Total number of feet of each size required.

(*g*) Identification marking, when not required (3.5).

(*h*) Levels of packaging and packing required (5.1 and 5.2).

(*i*) Special marking for shipment required (5.3.1).

6.3 **Hydrostatic pressure.** For sizes not listed in table II, internal hydrostatic test pressure for thin, hollow cylinders shall be as follows:

$$P = \frac{2tS}{D-.8t}$$

in which:

P = Safe internal pressure per square inch.
D = Outside diameter, in inches.
S = Allowable unit stress of the material, 8,000 pounds per square inch.
t = Wall thickness, in inches.

6.4 The chemical and mechanical property requirements of this specification are similar to those of ASTM B43-58.

6.5 **Transportation description.** Transportation description and minimum weight applicable to this commodity are:

Rail:

Pipe, brass, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:

Pipe, brass, not otherwise indexed.
Motor volume minimum weight 30,000 pounds.

6.6 **Supersession data.** This specification includes the requirements of Military Specification MIL-P-2780 dated September 4, 1951.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication, or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Copies of this specification may be purchased for nts each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:22Z
Check the source to verify that this is the current version before use.